

ProbOS — Month 8 Pre-Plan

Medical Device Domain (Part 1: Research Gate)

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Month goal

Determine, via real research BEFORE any model code, whether medical device battery safety can be genuinely validated.

The real, open question, stated up front

Implantable device batteries typically use lithium-iodide or lithium-silver-vanadium-oxide chemistry – NOT the lithium-ion chemistry `BatteryModel2Cell` already models. Unlike automotive, medical device may require a genuinely new model.

Week 30 – Research gate

- Search for real, citable implantable device battery safety data, matching the pharma model's Day 1 discipline
- Confirm or reject the chemistry-transfer assumption with real sources

Study materials for this week:

- FDA guidance documents on implantable device battery safety
- ISO 14708 (active implantable medical devices)
- A survey paper on implantable-device battery electrochemistry (post-2015, IEEE/electrochemistry journals)

Week 31 – Decision point

- If real data found: design the correct chemistry-specific model
- If not found: document honestly, pivot to Month 10's research gate early

Exit criteria

An honest, evidence-based answer – not a model built on an assumed chemistry transfer.